

Mode 8 — Baseline-BFP

DP16 (16×16) → DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents

Step 1 — DP16 (16×16)

$$\text{DP16 (16} \times \text{16)} = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (16×16)

$$\text{DP16 (16} \times \text{16)} = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8 (16} \times \text{16)}} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8 (16} \times \text{16)}}$$

Step 3 — DP8 (16×8)

$$\text{DP16 (16} \times \text{16)} = 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}}$$

Step 4 — DP8 (16×4)

$$\begin{aligned} \text{DP16 (16} \times \text{16)} = & 2^{12} \underbrace{\sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}} \\ & + 2^{12} \underbrace{\sum_{i=8}^{15} a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}} \end{aligned}$$

Step 5 — DP8 (8×4)

$$\begin{aligned} \text{DP16 (16} \times \text{16)} = & 2^{20} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{20} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{16} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^{16} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} \\ & + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + 2^{12} \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} \\ & + 2^8 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + 2^8 \underbrace{\sum_{i=8}^{15} a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \\ & + 2^4 \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=8}^{15} a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \end{aligned}$$

Step 6 — Block exponents (one exponent block per DP8 (8×4))

$$\begin{aligned} & \left(2^{20} \sum_{i=0}^7 a_i^{hi} b_i^{hh} + 2^{16} \sum_{i=0}^7 a_i^{hi} b_i^{hl} + 2^{12} \sum_{i=0}^7 a_i^{lo} b_i^{hh} + 2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{lh} \right. \\ & \quad \left. + 2^8 \sum_{i=0}^7 a_i^{lo} b_i^{hl} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{ll} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{lh} + \sum_{i=0}^7 a_i^{lo} b_i^{ll}, E_0 \right), \left(\dots i \in [8, 15] \dots, E_1 \right) \longrightarrow \left(M, E \right) \end{aligned}$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; DP16 (16×16) = $M \cdot 2^E$.

Step 7 — Result

$$\begin{aligned}
M = & 2^{-\delta_0} \left(2^{20} \sum_{i=0}^7 a_i^{hi} b_i^{hh} + 2^{16} \sum_{i=0}^7 a_i^{hi} b_i^{hl} + 2^{12} \sum_{i=0}^7 a_i^{lo} b_i^{hh} + 2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{lh} \right. \\
& + 2^8 \sum_{i=0}^7 a_i^{lo} b_i^{hl} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{ll} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{lh} + \left. \sum_{i=0}^7 a_i^{lo} b_i^{ll} \right) \\
& + 2^{-\delta_1} \left(2^{20} \sum_{i=8}^{15} a_i^{hi} b_i^{hh} + 2^{16} \sum_{i=8}^{15} a_i^{hi} b_i^{hl} + 2^{12} \sum_{i=8}^{15} a_i^{lo} b_i^{hh} + 2^{12} \sum_{i=8}^{15} a_i^{hi} b_i^{lh} \right. \\
& + 2^8 \sum_{i=8}^{15} a_i^{lo} b_i^{hl} + 2^8 \sum_{i=8}^{15} a_i^{hi} b_i^{ll} + 2^4 \sum_{i=8}^{15} a_i^{lo} b_i^{lh} + \left. \sum_{i=8}^{15} a_i^{lo} b_i^{ll} \right), \\
E = & \max(E_0, E_1)
\end{aligned}$$